Worksheet: Gains From Trade

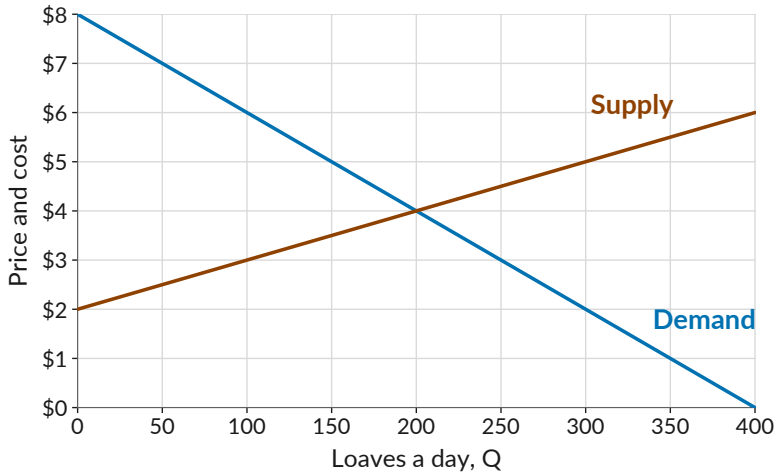
ECON 201

Definitions

- **Consumer surplus (CS)**: the sum of WTP $-P$ over every unit bought, the area between the demand curve and the price.
- **Producer surplus (PS)**: the sum of $P -$ MC over every unit sold, the area between the price and the supply curve.
- **Total surplus**: consumer surplus plus producer surplus, the gains from all the trades in the market. It is largest at the competitive equilibrium, where the last unit's WTP equals its MC.
- **Deadweight loss (DWL)**: the surplus lost on trades that would create surplus but do not happen.

A price cap on bread

The bakeries in town all sell the same loaf. The graph shows the demand and supply curves for bread. In equilibrium, 200 loaves a day sell at \$4 each, and consumer surplus is \$400 and producer surplus \$200.



1. The city council decides bread is too expensive and caps its price at \$3 a loaf. At \$3, how many loaves do the bakeries bake? How many do buyers want? How many loaves sell? Mark the price and the quantity sold on the graph.

2. Shade the consumer surplus and the producer surplus under the cap, and compute each.

3. Compared with the equilibrium, who gains from the cap and who loses? By how much?

4. Shade the deadweight loss and compute it. Which trades does it come from?

5. Is the outcome under the cap Pareto efficient? Explain in one sentence.